



Lesson Aim

Explain unique processes used to make white and sparkling wines.

INTRODUCTION

White wines can be made with either white or red grapes, though it is more commonly made from white. Skins, pips, and stems need to be removed after crushing, as they will add tannins and colour to wine. This is a key point of difference to red wines, which are made with skins, pips and even stems being left in to contribute tannins and colour.

Fermentation time is determined by the type of wine required:

- Sweet wine is produced by stopping fermentation before all of the sugar is converted to alcohol.
- Dry wine is produced by allowing all sugar to be converted to alcohol.
- Sparkling wine is made several ways; the most natural being to bottle the wine before fermentation is complete and allow fermentation to finish in the bottle.

HARVESTING GRAPES

As grapes ripen, the percentage of acid in fruit decreases and the percentage of sugar increases. Flavour and aroma strengthen as this process happens. The wine being produced is affected significantly by the stage at which they are harvested. Before harvesting white grapes, the fruits should be at 16 degrees brix or above with acid levels between 1 and 0.6

Under cold conditions, sugar levels might not ever rise as high as the winemaker wants, and acid levels might stay higher than desirable. At some point the winemaker might harvest grapes with low sugar and counter the issue by adding sugar to the grapes after pressing.

Wet, humid conditions close to harvest can also pose a problem. This moisture can promote excessive leaf growth reducing the normal ventilation that occurs as foliage thins. That can in turn raise humidity around fruit and increase diseases.

Hot dry conditions might pose less disease risk, and result in good sugar levels, but they can also reduce acid levels below what is required to make a good wine.

Minimise damage to berries Juices from broken berries can undergo enzymatic oxidation and spoil, more so if temperature is high. Oxidation is greatest at 30°C, but minimal at temperatures under 10°C. Picking grapes on cool days or at cooler times of the day helps to minimise this issue. Reducing the time between picking and crushing can also help. Sodium metabisulphite is sometimes dusted over grapes at the harvest site for added protection from unwanted microorganisms. If this is done, less chemical is needed later when sulphiting is done.

Grapes may also be left on a vine longer, to create higher sugar levels, making them suitable for producing a sweeter Spätlese style wine.

Harvesting can be by hand or using a mechanical harvester. Machine harvesting can greatly reduce picking costs for large scale operations but may not be viable for small vineyards. Mechanical harvesting is not appropriate to use with riper softer berries. ?

Avoid harvesting anything containing foreign materials or residues (e.g. pesticides and foliar fertilisers (even dust from a dust storm) can find their way into wine if the harvest is not “clean”).

MAKING THE WINE

Winemaking begins with producing the must, which is then fermented. The fermentation process is complex, and the knowledge and skill required to manage it well distinguish an exceptional wine from an average one.

Crushing the Grapes

Crushing harvested grapes involves breaking the skins to allow fermentation to start in a red wine; or to make pressing easier for a white wine.

Stalks are normally removed before crushing. Stems contain a lot of phenols which if left in the fermenting must, can result in a wine that is excessively bitter.

Modern crushing machines are designed to remove stalks as they crush the grapes.

Poorly designed de-stalking machinery can release too many phenols as they remove the stalks – resulting in a bitter taste. Well-designed machinery does not do this.

Crushing does release a quantity of juice, but the total quantity is far from *all* of the juice. Free running juice from crushing can be collected, drained off and fermented separately. This “free run” juice is considered optimum quality and is often used for producing better quality wines.

After crushing, grapes used for white wines are usually moved to pressing immediately.

Percolytic enzymes (commercially available) may be added to the crushed grapes. These enzymes help cell walls break down, thus increasing the volume of free flow juice. They can result in a greater quantity of wine, but overuse may reduce quality. Quantities of these enzymes added are recorded and listed among additives on the label.

Pressing extracts juice beyond what crushing does and separates the must (extracted liquid) from the skins. Hypothetically, if free run juice from crushing extracted 30% of the juice, a light pressing might extract a further 40% of high-quality juice. Extra, heavier pressing beyond that can extract the remaining juices, but that final pressing will be of a lesser quality.

Excessive pressing should be avoided, as it can extract high levels of phenols from crushed skins and pips, contaminating juice from that last pressing.

Managing the Must

The liquid obtained after pressing is called “must”, which you have already learned in earlier lessons. The must is usually treated with sulphur dioxide and allowed to settle for 24 hours at a low temperature (ideally around 0°C; never above 10°C). After settling the clearer juice is removed leaving sediment behind. Settling may be carried out in sealed, cooled containers with a layer of carbon dioxide injected above the surface to remove contact with air.

Some winemakers add a small amount of bentonite (1.5gm per litre) to help sediment settle. Some winemakers however believe that bentonite use during must settling can damage the ultimate quality. After settling, a must is best stored for around 10 days before blending. The must needs to be stable over that period though, which it is, if it has been treated with sulphur dioxide is covered with a layer of carbon dioxide and is stored at a low temperature (around 5°C).

The extracted must (after settling) is tested for acid and sugar levels. If either is inappropriate, adjustments need to be made. The best way to achieve a desirable balance is by blending different musts. Adding a must with too much acidity to one with too little can create a more balanced must. High sugar musts can be blended with low sugar musts. If acid and sugar levels cannot be perfected by blending, they may need additives to adjust those levels.

Blending is sometimes done after fermentation as well.

Once acid and sugar levels are right, the must can be moved to the fermentation stage.



Temperature management is more important for white wine fermentation than for red. The more volatile components of must are major contributors to subtle flavours and aromas in a wine. These include Acetic acids, aldehydes, esters, and higher alcohols. Such components are less stable at higher temperatures. White wines tend to have a more delicate flavour than red wines. To achieve that delicate quality, white wines are better to be fermented at a lower temperature, but to achieve that it is important to choose and use yeast that are more suited to lower temperatures.

Sealed fermentation tanks made from stainless steel (or other metals) which can be pressurised allow control over temperature through control over the rate at which carbon dioxide is released from the tank. When a must ferments in a tank at a steady rate the wine being produced is not able to be damaged very easily. It is sealed from contaminating microorganisms are controlled> Desirable yeasts dominate. Brix will be reducing by around 1 each day.

SPARKLING WINE

Sparkling wines generally start with a base wine, which might be either white, rose or red.

There are three main methods of creating sparkling wines:

- Charmat
- Champenois
- Carbonation

Charmat is the most widely used method; however, Champenois produces the best quality wines.

Carbonation involves injecting carbon dioxide into a base wine when it is ready for bottling. This is easier and cheaper than other methods, but once the sealed bottle is opened, it goes flat much faster than wines produced using the other methods.

The Charmat method involves adding a mix of sugar and yeast to a base wine, in order to restart fermentation. This is then placed into a sealed, pressurised tank. As fermentation proceeds, the carbon dioxide produced is held dissolved in the wine, unable to escape from the sealed container. The container must be able to withstand the increasing pressure. Once the process is finished, the wine is sterile filtered then bottled under pressure.

Champenois is the traditional champagne making method. It is more complicated and costly but will produce the best quality sparkling wines. It commonly involves six steps in the process, but winemakers do differ in the steps they take and what they do in each step. In that way different wine makers are able to strive to improve on their competition and develop different products.

The Champenois Method

The six steps in the process are:

1. Coupage (Blending)
2. Liqueur de Tirage (adding yeast and sugar)
3. Bottle Fermentation
4. Remuage (Clearing)
5. Degorgement (Recorking)
6. Sweetening and Final Corking

Coupage

Coupage involves blending base wines which may be made from just white or both white and red grapes. Traditionally there are various styles of sparkling wines that may be produced.

The Blanc du blanc style is made using 100% white grapes. Vintage champagnes are made using a blend of wines that are all produced in the same year.

Liqueur de Tirage

Liqueur de tirage is the process of adding cane yeast and a sweetener such as cane sugar, to the blended base wine. This stimulates fermentation to start again. The added sugar will feed the yeast supporting the secondary fermentation. Around 50% of added sugars are converted to carbon dioxide. It would be common to add around 25 gm of sugar per litre at this stage. A small amount of citric acid is also commonly added to help sucrose from cane sugar to convert to fructose and glucose.

Bottle Fermentation

Bottle fermentation involves placing the wine into appropriate bottles, sealing with a cork, and wiring the top to stop the cork popping out as pressure builds during the secondary fermentation. Secondary fermentation takes 3 to 6 months with bottles stored on their sides at an even temperature of 4-8°C. Bottles are periodically turned or shaken to stop the lees settling. Bottles may be left for years.

Remuage (Clearing)

Remuage is a process where bottles are gradually tipped from a horizontal position into a vertical position. As this happens, any remaining sediment settles on the cork. Originally this was done by securing bottles on a rack in a horizontal position, then gradually, over weeks, tilting the rack until the bottle neck and cork was at the bottom. This process has also been called riddling. This process is now mostly carried out using a machine.

Degorgement (Recorking)

Degorgement is a process that involves removing the cork and sediment that has settled on it. One method of doing this is to gently pry off the cork and allow pressure from inside the bottle to push out sediment; after which it is sealed with a fresh cork. Another technique involves freezing the cork and neck before removing the cork.

Sweetening and Final Corking

After disgorgement, the bottle needs to be recorked. At this point, a sweetener (e.g. sugar syrup) might or might not be added before recorking. If no (or little) sweetener is added, the sparkling wine will be called “brut”. If the wine has a medium amount of sweetener added it is called “sec” and if it has a lot more sweetener it is called “demi-sec”.

- Brut may have up to 1.5% added sweetener by volume
- Dry sec around 4%
- Demi sec around 8%
- Doux around 12%

